

Shankhanil Borthakur

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EDUCATION

University of Michigan, Ann Arbor

M.S. in Computer Science and Engineering

GPA: (expected)

Aug 2026 - May 2028

National Institute of Technology, Surat

B.Tech. in Computer Science (Minor), Electronics and Communication (Major)

GPA: 3.95/4

Aug 2019 - May 2023

PROFESSIONAL EXPERIENCE

Software Engineer 2, JPMorgan Chase

Aug 2023 – June 2026

- Led the engineering of a cloud-based distributed application that presently supports a **\$2B** annual trading business.
- Redesigned a high-throughput data pipeline to process **5M+** daily transactions, ensuring timely delivery of **SLA-critical** financial reports to regulatory bodies.
- Enhanced data warehousing capabilities by implementing a hot/cold data strategy leveraging Amazon S3 and **Redshift Spectrum**. Reduced infrastructure spending by **30%**.
- Architected AWS **multi-region** failover strategies to ensure minimal business downtime during regional cloud-service disruptions.

Software Engineer Intern, JPMorgan Chase

May 2022 - July 2022

- Designed a **Kinesis Firehose** streaming architecture to transform object-oriented trading data into relational schemas for Aurora RDS. This enabled **4X** scaling of trading analytics via simpler SQL aggregations and joins.

RESEARCH EXPERIENCE & PUBLICATIONS

Indian Institute of Technology Delhi - Implemented **Twin Augmentation** to address class imbalance in Covid-19 chest X-ray classification, improving model accuracy by **5%**.

Indian Space Research Organization - Developed a **geolocation algorithm** to map Mask R-CNN instance segmentation outputs to precise geo-coordinates in satellite imagery for remote sensing applications.

NER for Extracting Tourist Attractions from Textual Information: A BERT-BiLSTM-CRF Approach, *2023 IEEE 5th ICAECC*. 

Object Detection for Military Surveillance using YOLO Framework, *2023 IEEE 20th INDICON*. 

PROJECTS

Streak Fit | AI-Powered Virtual Fitness Platform 

- Led a **three-member team** to develop a fitness platform where multiple users can exercise via **video conferencing**, and a pose detection model provides real-time feedback.

Q&A Made Easy | Real-Time Q&A Moderation System using NLP 

- Chrome extension with a built-in NLP filter to remove **profane** and **repetitive** questions during online meetings. Integrated **speech-to-text** support and an admin dashboard for Q&A moderation.

Travelo | Smart Travel Planning using Named Entity Recognition 

- Web application to **parse** lengthy travel blogs, extract tourist destinations and generate an itinerary. It also plots an **efficient route** connecting the travel destinations on a map.

TECHNICAL SKILLS

Programming: Python, C++, Java, SQL

Tools & Technologies: Django, Spring, React, Docker, Kubernetes, GIT

Cloud & Data: Terraform, AWS, GCP, PostgreSQL, MongoDB, Redshift

ML / AI: LLMs, Agentic AI, RAG, Reinforcement Learning, NLP, OpenCV, TensorFlow, PyTorch

ACHIEVEMENTS & LEADERSHIP

- Gold Medalist**, ranked 1 out of 156 students in the Department of Electronics Engineering, NIT Surat.
- Accelerated Promotion** to Software Engineer 2 at JPMorgan, awarded to **top 5%** of the global cohort for impactful performance.
- Winner** of **7** hackathon awards, including annual competitions at the **University at Buffalo** and **Fordham University**.
- Microsoft Student Ambassador:** Mentored **50+** students in web development and organized **10+** programming workshops.
- Volunteered to develop a digital solution for a **non-profit** that automated **95%** of their manual data collection workflows.
- Organizer, DotSlash Hackathon: **Spearheaded technical operations**, developed event website, hosted **live sessions** on YouTube.